

Langevin Equation

$$m\ddot{x} = -\beta\dot{x} + A \rightarrow \begin{array}{l} \text{no external force field} \\ \text{drag force} \quad \text{stochastic force} \end{array}$$

'Work' equation:

$$m x \ddot{x} = -\beta x \dot{x} + x A$$

$$\frac{dx\dot{x}}{dt} = x\ddot{x} + \dot{x}^2 \Rightarrow x\ddot{x} = \frac{dx\dot{x}}{dt} - \dot{x}^2$$

$$x\dot{x} = \frac{1}{2} \frac{dx^2}{dt} \Rightarrow \frac{dx\dot{x}}{dt} = \frac{1}{2} \frac{d}{dt} \frac{d}{dt} x^2 = \frac{1}{2} \frac{d^2}{dt^2} x^2$$

$$\Rightarrow \frac{m}{2} \frac{d^2}{dt^2} x^2 - m \dot{x}^2 = -\frac{\beta}{2} \frac{dx^2}{dt} + x A$$

Average  over $N \rightarrow \infty$ random walking particles:

$$\begin{aligned} \left\langle \frac{d^2}{dt^2} x^2 \right\rangle &= \frac{1}{N} \sum_{n=1}^{\infty} \frac{d^2}{dt^2} x_n^2 = \frac{1}{N} \frac{d^2}{dt^2} \sum x_n^2 \\ &= \frac{d^2}{dt^2} \langle x^2 \rangle \end{aligned}$$

$$\langle \dot{x}^2 \rangle = \frac{kT}{m} \rightarrow \text{equipartition of trace ensemble}$$

$\langle xA \rangle = 0$ since forces due to random collisions
(for a group of particles near x , A just as likely to be positive as negative)

$$\Rightarrow \frac{m}{2} \frac{d^2}{dt^2} \langle x^2 \rangle - kT = -\frac{\beta}{2} \frac{d}{dt} \langle x^2 \rangle + 0.$$

$$\text{let } S = \frac{d\langle x^2 \rangle}{dt}$$

$$\Rightarrow \frac{m}{2} \dot{S} - kT = -\frac{\beta}{2} S$$

$$\Rightarrow \frac{m}{2} \dot{S} + \frac{\beta}{2} S = kT$$

$$\Rightarrow \dot{S} + \frac{\beta}{m} S = 2 \frac{kT}{m}$$

$$\text{RHS} = 0 \Rightarrow S = S_0 e^{-\frac{\beta}{m} t}$$

$$\text{particular solution} \Rightarrow S = C \Rightarrow \frac{\beta}{m} C = 2 \frac{kT}{m}$$

$$\Rightarrow C = \frac{2kT}{\beta}$$

$$\Rightarrow S = \frac{2kT}{\beta} + S_0 e^{-\frac{\beta}{m} t}$$

$t \rightarrow \infty \Rightarrow$ 'excess' drift damped out and just left with thermal fluctuations

$$S = \frac{2kT}{\beta} = \frac{d\langle x^2 \rangle}{dt}$$

over infinitesimal time τ (actually, macro $\tau \gg \tau_0$)

$$\Rightarrow \langle x^2 \rangle_{t+\tau} - \langle x^2 \rangle_t = \frac{2kT}{\beta} \tau$$

$$\begin{aligned} \langle [x(t+\tau)]^2 \rangle &= \langle [x^2(t) + 2x(t) \delta x_\tau + \delta x_\tau^2] \rangle \\ &= \langle x^2 \rangle_t + \langle \delta x^2 \rangle_\tau = \langle x^2 \rangle_{t+\tau} \end{aligned}$$

If τ is the mean time between collisions, then $\langle \Delta x^2 \rangle_\tau$ is the mean-square displacement per step

$$\Rightarrow \frac{\langle \Delta x^2 \rangle_\tau}{\tau} = D_x$$

$$\Rightarrow \boxed{D_x = \frac{kT}{\beta}}$$

$$\text{HE} \\ \frac{dp}{dz} = \rho g$$

$$\text{EOS} \\ p = \rho \frac{kT}{m}$$

$$\Rightarrow \frac{kT}{m} \frac{d\rho}{dz} = \rho g \rightarrow \text{thermal eqn.}$$

$$J = -D \frac{\partial \rho}{\partial z} \rightarrow -D \frac{d\rho}{dz} = v \rho$$

$$\frac{d\rho}{dz} = \frac{\rho g}{\frac{kT}{m}} \Rightarrow -D \frac{\rho g}{\frac{kT}{m}} = v \rho$$

$$\Rightarrow D = - \frac{kTv}{\rho g}$$

$$mg = \beta v \Rightarrow \frac{v}{g} = \frac{m}{\beta}$$

$$\Rightarrow |D| = \frac{kT}{m} \cdot \frac{m}{\beta} \rightarrow \text{valid for } \boxed{g \rightarrow 0} \text{, for finite } \beta. \\ \mu \neq m \text{ in general.}$$

$$\text{Stokes} \Rightarrow \beta = 6\pi \mu a \quad \text{viscosity.} \\ \text{sphere radius}$$